

One Object, Three Descriptions: Axes, Images and Grids of a Jacquard Pattern System

Anibal Edelberto Amiot

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Abstract

A Jacquard pattern system has been described in three separate ways: as sixteen families of straight axes on a 12×12 square combined by a parity rule; as sixty three-tint images from which 512 grids are built by a six-bit selection; and as level sets of a square-plate eigenmode. The images were generated and classified before the axes were formalised. We show, by exhaustive check on the published data and not by sampling, that the three descriptions present one finite structure, in a sense made precise in §1.

The central statement is that **every one of the sixty images is a parity figure of axes**: forty-eight cells of *lines*, the same for all sixty and equal to the nodal set of the degenerate guided plate mode $\cos(4\pi x/a) - \cos(4\pi y/a)$; ninety-six cells carrying the parity of a union of the four families of level T_0 ; and one of four tint permutations, which are the four *natures* of the system. The four T_0 families are a basis of a space $\mathbb{F}_2^4$ that the parity map embeds injectively, so the fifteen families of the loom are its fifteen non-zero vectors, and the involutions previously observed between natures are now their definition. The half-shift is then a linear form on $\mathbb{F}_2^4$, and the half-shift theorem of the companion note on grids — exactly eight families of fifteen admit unified pairs — is the count $2^4 - 2^3$ of vectors off its kernel.

Two further results fix the status of the construction. The code spanned by the sixteen families is a $[1152, 13]$ binary code of minimum distance 288: the integrity check that located two corrupt plates is a linear code that detects 287 wrong regions and corrects 143. And, on levels, the doubling homothety that organises the families into generations terminates without cycles if and only if the period is 2^a or $3 \cdot 2^a$, and has a non-trivial fixed level if and only if 3 divides the period: 12 is the smallest period meeting both criteria with at least two generations.

Nothing here is measured on a plate; the plate statements are exact for guided edges only, and the one interferometric prediction is recalled from the companion note, not used. All scripts and data are public.

1. Three corpora

The system in question was designed by hand, over several years, as a grammar of woven patterns. It has been formalised in three notes, each verified region by region against the published vector drawings.

The axes [A]. Sixteen families of straight lines — horizontal, vertical, and the two diagonal directions — on a 12×12 square, indexed by a level $T_0, \dots, T_3$ and a name among YIN, YIN MUT, YANG, YANG MUT. A subset of families, called a *chord*, produces a two-colour figure by a parity rule. The construction is reversible, the reading cell is a parameter, the families are closed under the doubling homothety, and the figures carry an integrity check.

The images and the grids [G]. Sixty three-tint 12×12 images, organised as fifteen families of four natures, from which 1024 constructions $\Phi(n, A, B)$ give 512 distinct grids under a six-bit selection. A half-shift criterion — translating a grid by half its period reproduces the exchange of its two layers — selects eight families of fifteen; the edge rule of a cube retains four; a trichotomy governs the crossed pairs.

The plate [P]. The sixteen axis families are level sets of the single function $\varphi(x) = \cos(\pi x/3)$ at the seven angles $0^\circ, 30^\circ, \dots, 180^\circ$; the mutation is the reflection $\theta \mapsto 180^\circ - \theta$; φ is an exact eigenfunction of the square Kirchhoff plate with guided edges, and the degenerate combinations $\cos(4\pi x/a) \pm \cos(4\pi y/a)$ have for nodal sets two of the diagonal families.

Independence of the corpora. The identification below is not a consequence of a common formalisation. The notes were written in the order [G], [A], [P]: the sixty images of [G] were generated, and classified into fifteen families and four natures, before the axis catalogue of [A] existed, and [G] makes no use of axes. The axes were extracted afterwards from the plates of the treatise [5], not from the images. Both corpora descend from the same treatise, so a common origin is not excluded — it is what the paper explains; what is excluded is that the conventions used to build one corpus were chosen to fit the other. The question this paper answers is whether [G] can be derived from [A]. It can, and the derivation passes through [P].

What “one object” means. By *one object* we mean this, and no more: a single finite structure — the 48-cell set L of §2, the space $\mathbb{F}_2^4$ of chords of the four T_0 families, and four tint permutations — of which each corpus is a presentation, region by region, on the published data. The claim is about finite, published data. Accordingly Theorem 1 is labelled a *finite-corpus identification*: its proof is an exhaustive check, and it says nothing about images that were not drawn. Theorem 2 is a finite computation on the 72 band systems; Theorem 3 and the Lemma of §3 are proved by argument. §8 sorts every statement by status.

Place of this paper. The system has an origin, a structure, and three extensions, each with its own note, and this paper is the second. The ansate cross [C] is the origin: a cross-shaped constraint system that fills the order-6 squares from which the treatise’s patterns descend. This paper describes the structure common to the three corpora. [G] asks whether that structure closes on a cube; [P] reads the same lines as level sets of a plate mode; [K] uses the resulting finite vocabulary as the referent of an encoding. Each of them depends on this paper only through Theorem 1, and none is repeated here.

We fix conventions once. The square is $[0, 12]^2$ divided into 144 unit cells; C_1 is that subdivision, C_8 cuts each cell by its two diagonals and two medians into eight triangles, 1152 in all. An axis is a line of one of four *natures* — H, V, D^+ ($x + y$ constant), D^- ($y - x$ constant) — given by its signed offset e from the centre, in cells; diagonal offsets are $(x + y - 12)/2$ and $(y - x)/2$. The generating objects are not the drawn segments but the **band systems**, or *keys*: an orthogonal key is a nature and an offset $e \in (-6, 6]$, of period 12; a diagonal key is a nature and an offset taken modulo 6, since a lattice translation by $(12, 0)$ shifts the diagonal coordinate by 6, and it is drawn on the square as the two parallel segments e and $e \pm 6$. There are 72 keys — 24 per orthogonal nature, 12 per diagonal nature — and the sixteen families are sets of keys, each key counted once. The **parity figure** of a set of keys on a subdivision $\mathcal{C}$ colours a region dark according to the parity of the number of lines of the systems that separate its centroid from a fixed reference point, taken just inside the corner at the origin (the scripts use $(0.01, 0.01)$); since no figure is identified with its complement, this point fixes every weight quoted below. We write $\Pi_{\mathcal{C}}(A)$. No identification of a figure with its colour exchange is made: on C_8 the all-dark figure is not in the span of the keys, so the exchange is not available, and on C_1 it is, which is a relation and not a convention. Under this rule the sixteen families reproduce the published plates region by region [A]; the inventory of *drawn segments* is 94, of which 30 lie in two families, but it is not the generating set, and counting it as one would double the two diagonal lines of T_0 YANG MUT and annul them.

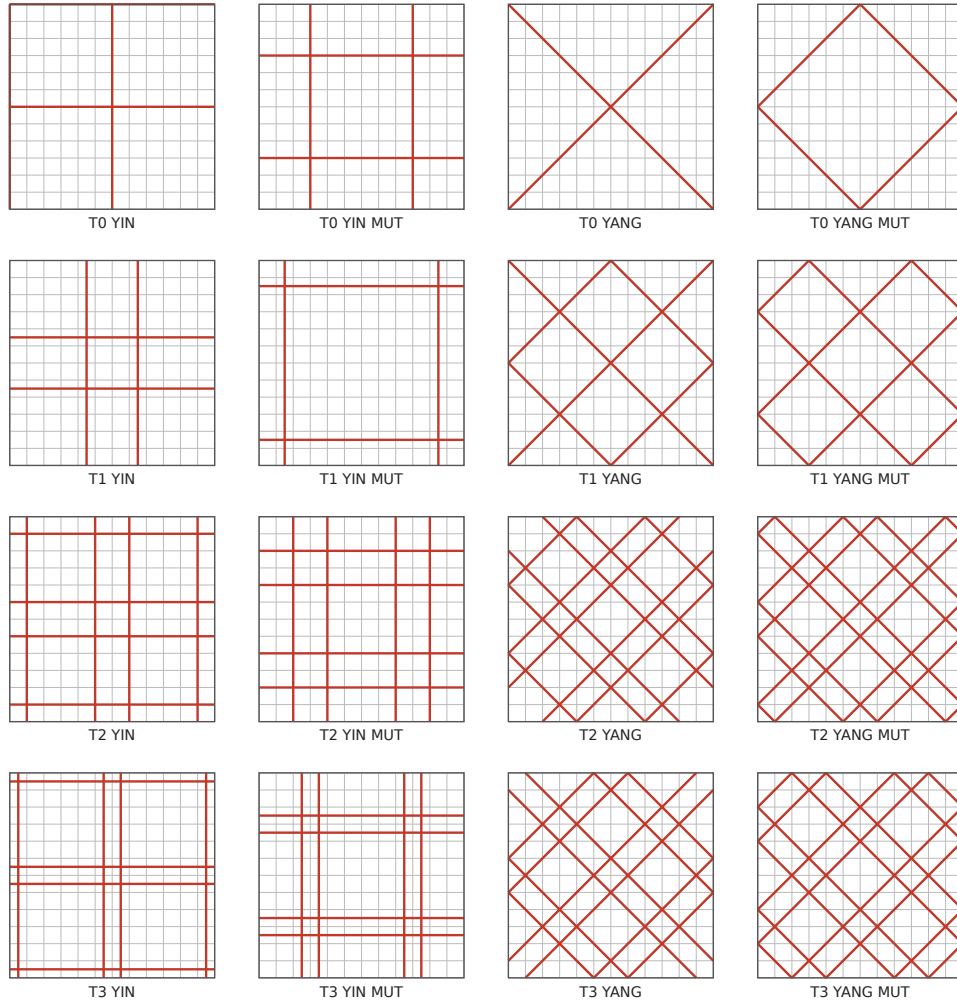


Figure 1. The sixteen families of axes, as drawn on the square: orthogonal (YIN) on the left, diagonal (YANG) on the right, levels T0 to T3 from top to bottom. The edge lines of T0 YIN coincide with the border.

For a chord, A is the union of the keys of its families; for the linear algebra of §5 we use the symmetric difference, under which a key present in two families cancels. The two semantics agree on all chords of T_0 families, whose key sets are pairwise disjoint.

2. The images are parity figures

Let L be the set of cells whose centre lies on an axis of T_1 YANG, that is, on one of the six diagonal lines of offset 0 or ± 3 in either sense. L has 48 cells. Let $\mathcal{B} = \{T_0 \text{ YANG}, T_0 \text{ YANG MUT}, T_0 \text{ YIN}, T_0 \text{ YIN MUT}\}$; its four members are pairwise disjoint sets of axes.

Theorem 1 (finite-corpus identification). *Every one of the sixty published images of $[G]$ is of the form*

$$I = (L, \Pi_{C_1}(F), \tau),$$

where F is a non-empty subset of $\mathcal{B}$, the 48 cells of L carry one tint, the 96 cells off L carry the two-colour figure $\Pi_{C_1}(F)$ in the two other tints, and τ is one of four tint assignments:

nature	cells of L	figure $\Pi(F)$	dark / light
YANG	O	V / M	
YANG MUT	O	M / V	
YIN	M	O / V	
YIN MUT	M	V / O	

The correspondence $F \leftrightarrow$ family is a bijection between the fifteen non-empty subsets of $\mathcal{B}$ and the fifteen families of the loom: the four BASES families are the singletons, PAR2 the six pairs, PAR3 the four triples, YINYANG the whole.

Proof. Exhaustive check on the sixty published images, cell by cell, with no exception (verif_images_axes.py). ◻

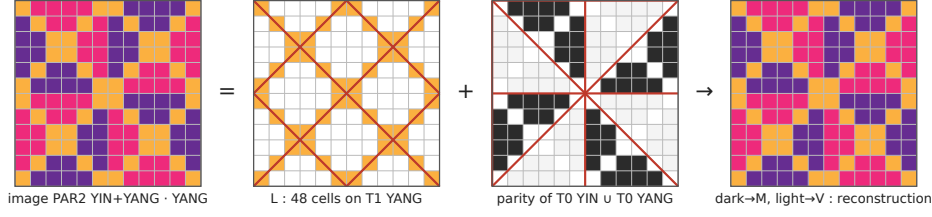


Figure 2. Theorem 1 on one image: the image of family PAR2 YIN+YANG, nature YANG, equals the 48 cells of L in orange, plus the parity figure of T_0 YIN $\cup$ T_0 YANG on the other 96 cells, dark to M and light to V ; the reconstruction matches cell for cell.

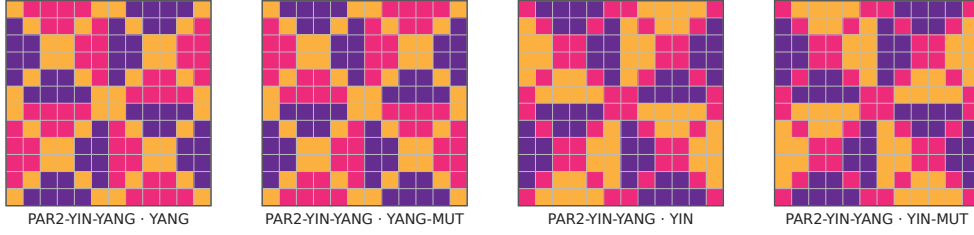


Figure 3. The four natures of one family (PAR2 YIN+YANG) are four tint permutations of the same lines-and-parity structure: YANG, YANG MUT (V and M exchanged), YIN ($V \rightarrow O$, $M \rightarrow V$, $O \rightarrow M$), YIN MUT (V and O exchanged).

Three remarks.

The natures are tint permutations. YANG MUT is YANG with V and M exchanged; YIN MUT is YIN with V and O exchanged; YIN is YANG under the cycle $V \rightarrow O$, $M \rightarrow V$, $O \rightarrow M$. All three hold on all fifteen families. The first two are exactly the involutions π recorded as fact (F5) in [G] — $V \leftrightarrow M$ fixing O for the pair (yang, yang-mut), $V \leftrightarrow O$ fixing M for (yin, yin-mut). They were observed there; here they are constructed.

The lines are a plate object. By Theorem 3 of [P], L is the nodal set of $\cos(4\pi x/a) - \cos(4\pi y/a)$, the difference of the two degenerate guided modes $(4, 0)$ and $(0, 4)$. Every image of the loom therefore carries, in its third tint, the nodal lines of one plate mode, and in its two other tints the parity of axes of level T_0 — the centre lines and the diamond.

The third remark deserves to be a statement, because the rest of the paper leans on it. Identify a subset $F \subseteq \mathcal{B}$ with its indicator vector in $\mathbb{F}_2^4$, coordinates in the order YANG, YANG MUT, YIN, YIN MUT, and let V_L be the space of two-colourings of the 96 cells off L , with 1 the all-dark colouring.

Proposition 1. *The map $\Phi: \mathbb{F}_2^4 \rightarrow V_L$, $F \mapsto \Pi_{C_1}(F)|_{C_1 \setminus L}$, is linear and injective, and its image meets the line $\{0, 1\}$ only in 0. Hence the fifteen non-zero chords of T_0 give fifteen figures that are pairwise distinct even up to colour exchange, and the fifteen families of the loom are the fifteen non-zero vectors of $\mathbb{F}_2^4$.*

Proof. Linearity: the members of $\mathcal{B}$ are pairwise disjoint sets of keys, so union and symmetric difference agree and $\Pi(F \cup F') = \Pi(F) \oplus \Pi(F')$. Injectivity and the second assertion are a rank computation over $\mathbb{F}_2$: the four basis figures have rank 4 on the 96 cells, and rank 5 with 1 adjoined (espace_F2_4.py). ◻

The count $15 = 2^4 - 1$ of families in [G] is thus not an inventory but a dimension, and the four natures act on a fixed image of Φ by tint permutation only.

3. The half-shift theorem, derived

[G] builds grids from two images A, B of one family by a six-bit word n selecting, level by level of a fixed map, which image supplies each cell, and calls a pair *unified* when $B = A \circ \sigma$, σ the translation by $(6, 6)$ modulo 12. Its Theorem 2 states that unified pairs exist in exactly eight families of fifteen, those containing exactly one base of yang type, and that the pair is then (yang, yang-mut) or (yin, yin-mut).

With Theorem 1 this is a statement about three sets of lines.

Lemma. *On the torus, σ maps each of the five axis sets L , T_0 YANG, T_0 YANG MUT, T_0 YIN, T_0 YIN MUT to itself. Consequently, for every $F \in \mathbb{F}_2^4$,*

$$\Pi(F) \circ \sigma = \Pi(F) \oplus \varepsilon(F) \mathbf{1}, \quad \varepsilon(F) = F_{\text{YANG}} + F_{\text{YANG MUT}} \pmod{2},$$

so that σ acts on figures through the linear form $\varepsilon: \mathbb{F}_2^4 \rightarrow \mathbb{F}_2$, equal to 1 on the two yang-type basis vectors and 0 on the two yin-type ones.

Proof. L is the set of lines $x + y \in \{6, 12, 18\}$ and $x - y \in \{0, \pm 6\}$, invariant under $(x, y) \mapsto (x + 6, y + 6)$ modulo 12. The same holds for the centre cross $\{x = 6, y = 6\}$, the diagonals $\{x = y, x + y = 12\}$, the square $\{x, y \in \{3, 9\}\}$ and the diamond $|x - 6| + |y - 6| = 6$. A parity figure is, up to complement, the unique two-colouring whose colour changes exactly across the lines of its axis set; so $\Pi(F) \circ \sigma$ is $\Pi(F)$ or its complement, and one point decides which. For T_0 YIN the quadrant checkerboard is invariant; for T_0 YIN MUT the centre and the corner are both “inside-inside” or “outside-outside” and have the same colour; for T_0 YANG the top wedge is sent to the left wedge, of opposite colour; for T_0 YANG MUT the centre, inside the diamond, is sent to the corner, outside. So ε takes the stated values on the basis, and it is linear because Π is (Proposition 1) and σ commutes with $\oplus$. $\square$

Corollary (Theorem 2 of [G]). *Since the mutant of yang is yang with the figure complemented and the lines fixed, $I(F, \text{yang}) \circ \sigma = I(F, \text{yang mut})$ if and only if F has exactly one yang-type member; otherwise $I(F, \text{yang}) \circ \sigma = I(F, \text{yang})$. The same holds for yin. The unified families are therefore the vectors with $\varepsilon(F) = 1$, the complement of the hyperplane $\ker \varepsilon$: there are $2^4 - 2^3 = 8$ of them, namely $\{Y\}$, $\{YM\}$ and their unions with any subset of the two yin-type bases; the $2^3 - 1 = 7$ non-zero vectors of $\ker \varepsilon$ are the invariant families.*

Verified on the data: for the eight families the yang image translated by $(6, 6)$ is the yang-mut image, cell for cell, and the yin image is the yin-mut image; for the seven others each image is invariant (demi_decalage.py, and espace_F2_4.py for the identity on all fifteen F). This is the whole content of Theorem 2 of [G], which was obtained there by enumeration of 1024 constructions; the numbers 8 and 7 are now a hyperplane and its complement.

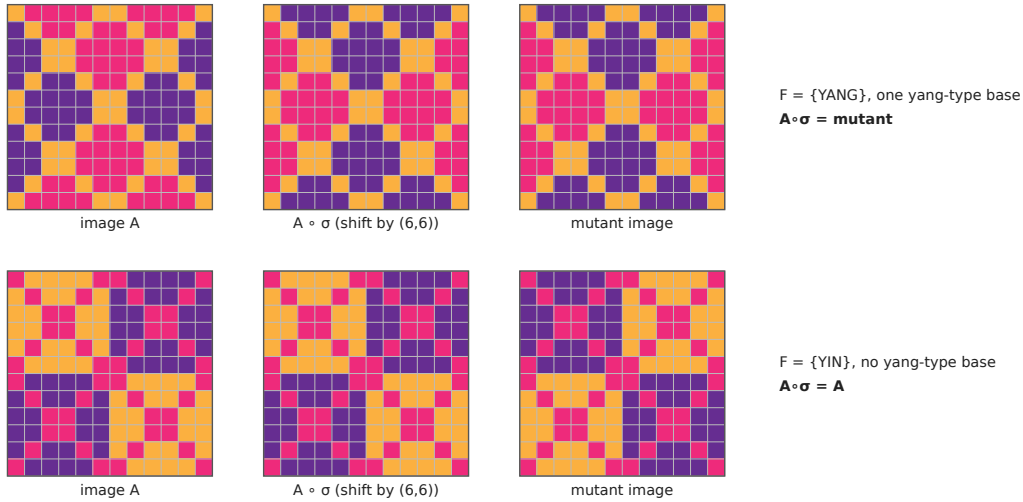


Figure 4. The Corollary of §3 on two families. Top, $F = \{\text{YANG}\}$: translating the yang image by $(6, 6)$ gives the yang-mut image. Bottom, $F = \{\text{YIN}\}$: the same translation leaves the image invariant.

The edge rule on the cube (Theorem 3 of [G]: a grid dresses the six faces if and only if its family contains the base yang, four families of eight) and the trichotomy on crossed pairs (Theorem 4) remain, at the time of writing, results verified by exhaustive search. Theorem 1 makes them statements about which of the two centre figures — the diagonals through the centre, or the diamond — accompanies L ; we have not derived them and do not claim to.

4. The plate reading, in the form needed

We recall from [P] only what is used. The eight orthogonal families are the complete level sets of $\varphi(x) = \cos(\pi x/3)$ at $\theta = 0^\circ, 30^\circ, \dots, 180^\circ$, read as positions $p = 6 + e$, the level 90° being split into T_1 YIN and T_1 YIN MUT; the eight diagonal families are level sets of the same function in the diagonal coordinate — as band systems all eight are complete, T_2 YANG and T_3 YANG MUT being the same level pair, as are T_3 YANG and T_2 YANG MUT (§5); the mutation is $\theta \mapsto 180^\circ - \theta$.

Exact, in the guided model. φ is an exact eigenfunction of the square Kirchhoff plate with all four edges guided (zero slope, zero shear), the mode $(4, 0)$, and the degenerate combinations $\cos(4\pi x/a) \pm \cos(4\pi y/a)$ are exact guided modes whose nodal sets are T_1 YANG MUT (sum) and T_1 YANG (difference). “Exact” means exact for guided edges, which have not been realised in a laboratory; for the free plate of Chladni’s demonstrations the cosine product is Colwell’s classical approximation [1, 3].

Predicted, not measured. Of the seven levels only 90° is nodal; the six others are iso-amplitude lines, which time-averaged holographic interferometry [2] would show at drive amplitudes in the ratio 1:1.155:2, on a rectangle where $(4, 0)$ is not degenerate. The ratio follows from the assumed interferometric response of the plate, not from the grammar; it is stated in [P] and nothing below depends on it.

What Theorem 1 adds is that the loom’s images use exactly two of these objects: the nodal set of the difference mode, and the parity of the 0° and 180° levels. The four generations $T_0, \dots, T_3$, the doubling homothety and the five other levels do not enter the images; they enter the *plates* of the treatise, from which the axes were extracted, and §7.

5. The integrity check is a linear code

Let $V_{\mathcal{C}}$ be the $\mathbb{Z}/2$ -vector space of figures on $\mathcal{C}$, and let $\lambda: \{\text{keys}\} \rightarrow V_{\mathcal{C}}$ send a key to the parity figure of its band system, so that $\Pi_{\mathcal{C}}$ of a symmetric difference of key sets is the sum of the λ ’s.

The code is not a second subject of this paper but a third presentation of the same structure: the figures of §2 seen as words. The statement collects Theorems 6 and 7 of [A] and adds the reading on C_1 ; it is restated because §2 and §7 lean on it.

Theorem 2 (finite computation on the 72 keys). *On C_8 : the two boundary keys, H and V at $u \equiv 0$, have $\lambda = 0$, and the remaining 70 keys are linearly independent, so λ is injective on them and a figure determines its set of band systems uniquely. The sixteen family figures span a subspace of dimension 13; the three relations are*

$$T_0 \text{ YANG} \oplus T_0 \text{ YANG MUT} \oplus T_1 \text{ YANG} = 0, \quad T_2 \text{ YANG} \oplus T_3 \text{ YANG MUT} = 0, \quad T_3 \text{ YANG} \oplus T_2 \text{ YANG MUT} = 0.$$

The code they span is a $[1152, 13]$ binary code of minimum distance 288, with a unique word of minimum weight, $T_0 \text{ YANG MUT} \oplus T_1 \text{ YIN}$ — the diamond against the inner square of offsets ± 1.5 . Under the union semantics of chords, the 65 535 chords produce exactly the $2^{13} - 1 = 8191$ non-zero words and never the zero word; under symmetric difference, seven chords — the non-zero elements of the relation space, of sizes 2, 2, 3, 4, 5, 5, 7 — give the empty figure, and the words are 8192.

On C_1 the test point of a cell is its centre, and 48 of the 72 keys pass through cell centres — every diagonal key, and every orthogonal key at a half-integer offset. An axis through a centre does not separate that cell: the cell closes the figure continuously, on the side of increasing coordinate, the same all along the axis ([A], `regle_parity.py`: $\lfloor (u - c) / 12 \rfloor \bmod 2$; `codes_cles.py` places ties on the same side). This rule is part of the definition of C_1 , and every number below depends on it. Under it the two boundary keys are again empty, and the two keys at offset $-5\frac{1}{2}$ put every cell on one side, so their figure is uniform — empty without anchoring, all-dark when parity is counted from the corner point, which lies below their line; the keys have rank 43 and the families rank 12 in both readings. The fourth relation on C_1 is that the eight orthogonal families sum to zero — their rank drops from 8 to 7, and $5 + 7 = 12$; their 48 keys are pairwise disjoint, so the relation holds under union as well as under symmetric difference, and that chord is the one chord whose figure is empty under union. Separately, and unlike on C_8 , the all-dark figure lies in the span on C_1 , its shortest expression being $T_1 \text{ YANG} \oplus T_1 \text{ YANG MUT} \oplus T_2 \text{ YANG} \oplus T_2 \text{ YANG MUT}$. The minimum distance is 36.

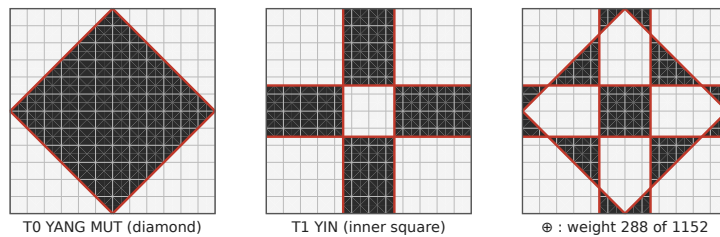


Figure 5. The unique minimum-weight word of the family code on C_8 : the parity of T_0 YANG MUT (the diamond), the parity of T_1 YIN (the inner square), and their sum, which differs from zero on exactly 288 of the 1152 triangles. Parity is counted from the reference point at the origin corner.

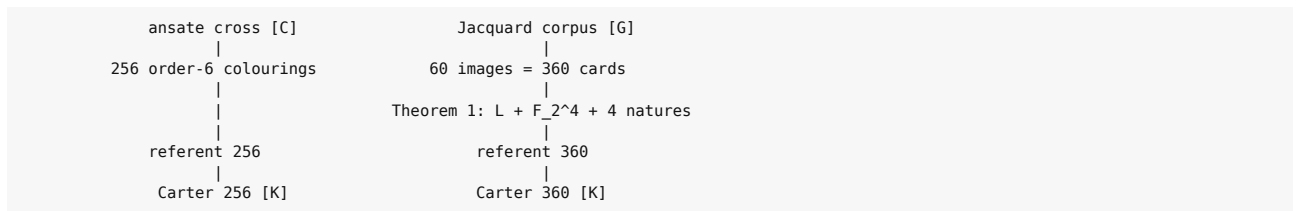
Consequences. The dimension 13 is not only computed; it can be read off the levels of §4. Across the diagonals the level of an axis of offset e is $\cos(\pi e/3)$, well defined modulo the period 6, and the eight diagonal families occupy five distinct sets of signed levels: 0° (T_0 YANG), 180° (T_0 YANG MUT), 90° (T_1 YANG MUT), $\{60^\circ, 120^\circ\}$ (T_2 YANG and T_3 YANG MUT) and $\{30^\circ, 150^\circ\}$ (T_3 YANG and T_2 YANG MUT), with T_1 YANG the union of the first two. The diagonal families therefore have rank 5; the eight orthogonal families are independent on C_8 and have rank 8; and $5 + 8 = 13$, the three relations being exactly the three diagonal families that repeat what is already spanned. The first relation is trivial (T_1 YANG is the disjoint union of the two T_0 diagonal families). The second and third say that T_2 YANG and T_3 YANG MUT are the *same figure*, and likewise T_3 YANG and T_2 YANG MUT: the two families that, as drawn, omit their corner segments (§4) are, as band systems, complete level sets, and the level-set identification of [P] is exact for all sixteen. The reversibility of chords (Theorem 5 of [A], by exhaustive sweep) is the restriction to unions of families of the injectivity on band systems (Theorem 7 of [A]), valid for every set of keys. The integrity check of [A] — “a figure that is the exclusive-or of no two others is foreign” — is membership in a code of distance 288: any corruption of at most 287 triangles is detected, and any of at most 143 is corrected by nearest-word decoding. At exactly 288 there is a hole: a corruption of that size can land on another word of the code and pass unseen, and the uniqueness of the minimum-weight word says precisely which one — the diamond exchanged against the inner square. A plate whose subdivision itself is broken, with regions missing or overlapping at extraction, has no Hamming distance to any word, and is rejected before decoding rather than corrected by it. The drop from rank 70 to 43 on passing from C_8 to C_1 is the quantitative form of the statement in [A] that the reading cell is a parameter.

A word on the control. The number 18 431 sometimes quoted alongside these counts ([A], after Theorem 7) is the number of distinct *unions of drawn segments* over the 65 535 chords, and a parity map that returns 18 431 distinct figures is one that does not identify a segment with its period-6 translate; it reproduces neither the plates nor the relations above. The figures are $8191 = 2^{13} - 1$, and that count is itself the statement that the family code has no other relation than the three listed.

6. From classification to referents

The classification is not only descriptive. It turns the corpus into a finite vocabulary of named configurations — a vector $F \in \mathbb{F}_2^4 \setminus \{0\}$, a nature, a level of the six-level map of [G] — and a finite named vocabulary can serve as the *referent* of a grammar of transformation. This has been done twice, and the two referents should not be confused.

The *referent 256* comes from the origin of the chain: the 256 admissible colourings of the order-6 squares of the ansate cross [C], two-colour *6imes6* blocks with which the Carter encoding [K] writes a message in a *90imes90* grid of 225 blocks. The *referent 360* comes from the corpus of this paper: the 360 cards of the sixty images, from which the Carter 360 implementation retains a table of 294 forms in three colour classes instead of two, written in a *180imes180* grid of 225 blocks of *12imes12*, the form, colour and orientation of each block being chosen by a grammar derived from a key [K].



Carter is thus a use of the structure, not its origin: Theorem 1 was not needed to build it, and nothing about the encoding is claimed here. Its construction and its limits belong to [K].

7. Why 12 is the smallest admissible period

Generalise the vocabulary to a period N : positions on the half-integer grid $p \in \{0, \frac{1}{2}, \dots, N - \frac{1}{2}\}$, the function $\cos(4\pi p/N)$, hence levels $\cos(2\pi j/N)$ for $j = 0, \dots, N/2$. We work on *levels*, that is, we identify an offset e with $-e$: every family of the construction is symmetric under $e \mapsto -e$, so the homothety acts on families through its action on levels. The hypothesis is not idle — without it, $N = 12$ has the cycle $4 \mapsto -4 \mapsto 4$ — and with it the doubling homothety of [A], $e \mapsto 2e$, doubles the angle: $j \mapsto 2j \pmod{N}$, modulo sign.

Theorem 3. *On levels (offsets identified with their opposites), write $N = 2^a M$ with M odd. The descent under doubling terminates without cycles if and only if $M \in \{1, 3\}$. A fixed level other than 0° exists if and only if $3 \mid N$,*

and it is 120° . The depth of the descent — the number of doublings before every level reaches a fixed one — is a .

Proof. After a doublings every j lies in the subgroup $2^a \mathbb{Z}/N \cong \mathbb{Z}/M$, on which doubling is a bijection; its orbits modulo sign are all fixed points if and only if $2j \equiv \pm j \pmod{M}$ for all j , i.e. $M \mid 3j$ for all j , i.e. $M \in \{1, 3\}$. A non-zero fixed level satisfies $3j \equiv 0 \pmod{N}$, hence exists iff $3 \mid N$, and is $j = N/3$. ◻

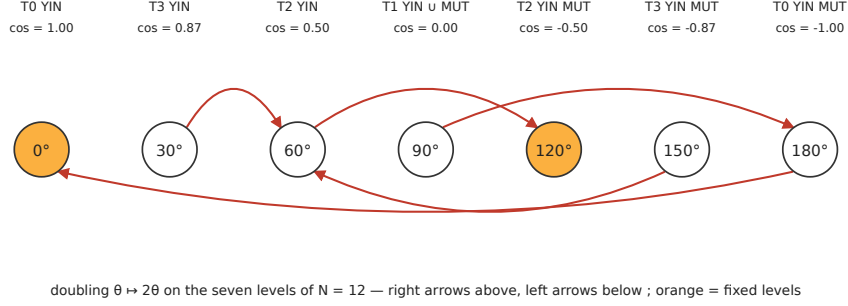


Figure 6. The doubling map on the seven levels of $N = 12$. Two levels are fixed, 0° and 120° ; every other level reaches one of them in at most two steps, which is the depth $a = 2$ of Theorem 3.

N	levels	terminates	depth	fixed levels
6	4	yes	1	$0^\circ, 120^\circ$
8	5	yes	3	0°
10	6	no ($72^\circ \leftrightarrow 144^\circ$)	—	0°
12	7	yes	2	$0^\circ, 120^\circ$
16	9	yes	4	0°
18	10	no	—	$0^\circ, 120^\circ$
24	13	yes	3	$0^\circ, 120^\circ$
48	25	yes	4	$0^\circ, 120^\circ$

So, relative to three criteria fixed in advance — the descent terminates, it possesses a non-trivial fixed level, and it has at least two generations before the fixed core — 12 is the smallest admissible period: its fixed level is the 120° of T_2 YIN MUT and T_2 YANG, and its two generations are $T_3 \rightarrow T_2$ and $T_1 \rightarrow T_0$. The criteria are those of the construction as drawn: the treatise’s families are organised by doubling, they include a 120° family, and they come in four levels. The theorem does not say why a maker would want them; it says that, wanting them, the period 12 is forced. The admissible periods form the sequence $3 \cdot 2^a$: 6, with one generation and four levels; 12; 24, with three generations and thirteen levels; 48. The pure powers of two close as well but have only 0° for fixed level; every other even period has cycles, and the echo never descends. Whether the loom’s fifteen families have an analogue at $N = 24$ — with $2^4 - 1$ or $2^6 - 1$ members — is an open question the theorem makes precise.

8. What is established, and what is not

status	statements
Proved by argument	the Lemma of §3 and its Corollary; Theorem 3
Proved by finite rank computation	Proposition 1 (rank 4, rank 5 with 1); Theorem 2 (Gaussian elimination on the 72 band systems, full enumeration of 2^{16} words)
Verified exhaustively on the finite corpus	Theorem 1 (60/60 images, 15/15 families for each tint relation); the Corollary on the data (8/15 and 7/15, cell for cell)
Verified by search, not derived	Theorems 3 and 4 of [G]: the edge rule and the trichotomy. Theorem 1 restates them as questions about the diagonals and the diamond; the answers are known, the reasons are not
Exact in a model not realised	the guided plate of §4
Predicted, not measured	the fringe ratio 1: 1.155: 2 of [P], recalled in §4 and not used
Not claimed	anything about images not in the published corpus; anything about the encoding of §6

Not claimed either: depth. Each of the three descriptions is elementary. What this paper establishes is that they present one finite structure; that the identification is not built into the conventions of either corpus (§1); and that, given the criteria of §7, the period 12 is the smallest that admits it.

9. Reproducibility

Every number in this paper is printed by a script that reads the published data and reruns in under five minutes: `verif_images_axes.py` (Theorem 1 and the tint relations), `espace_F2_4.py` (Proposition 1 and the linear form ε), `demi_decalage.py` (the Corollary of §3), `codes_cles.py` (Theorem 2, on band systems), `harmonie_N.py` (Theorem 3), on the shared module `common.py` (parity map on C_1 and C_8 , $\mathbb{Z}/2$ elimination). Inputs: `catalogue-axes.json` (the sixteen families) and `referent_360_v3.json` (the sixty images); `catalogue-axes.json` is the file deposited with [A] and [P] (doi:10.5281/zenodo.22965031), the sixty images are those of [G] (doi:10.5281/zenodo.22862110). As a control, the parity map used here returns 8191 distinct non-empty figures on the 65535 chords under union, 8192 words under symmetric difference, the count of [A], and the three relations of Theorem 2. (The script lists eight chords of minimum weight on C_8 and thirty-two on C_1 : these are chords, not words — one word times 2^3 for the three relations, and two words times 2^4 — and the uniqueness of Theorem 2 is at the level of words. An independent implementation using a floor rule instead of the differential count from the corner reproduces every number.)

Code under AGPL v3; text, figures and data under CC BY 4.0. Context and chronology of the three notes: [R].

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